Fixed testing functions at singular C1 contacts

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A finite semidefinite construction needs testing functions that remain fixed as the supporting Bell inequality varies. We construct such a space for the reciprocal C1 contact family on an explicit source and testing rectangle. The ratio $H = h/\sigma$ is unrestricted, and either event weight D or β may vanish.

Two identities control the state couplings through the sum of complementary source weights. They give a fixed positive diagonal Gram matrix with uniformly small corrections. We then separate its source and testing coefficients by one scalar identity. Including every retained square gives 220 testing functions for $\lambda > 0$ and 157 for $\lambda \le 0$. Their union therefore supplies at most 377 functions for the complete real-state parent on this rectangle. The other source domains and projective testing charts remain necessary for a global pencil.

1 The rectangle and source formulas

Fix the rational constants

$$\varepsilon = 2^{-160}, \qquad \delta = 2^{-100}, \qquad c_0 = 2^{-8}, \qquad A_* = 2^{30}, \qquad \gamma = \frac{c_0}{7A_*} = \frac{1}{7\,2^{38}}.$$

The source domain consists of

$$0 < x, \sigma \le \varepsilon, \quad 0 < z \le 1, \quad 0 < b < 1, \quad r = bz < 1, \quad 0 < h < 1,$$

with the original strict gaps and the two nonnegative event coefficients specified below. Set

$$Q = 1 - b, \quad \theta = 1 - h, \quad H = h/\sigma, \quad e = xh/\sigma, \quad E = 1 - e > 0, \quad k = \sqrt{x}, \quad q = \sqrt{z},$$

$$K = 1 + x, \quad R_s = 1 + xr, \quad M = R_s - xz\theta > 0, \quad J = 1 + xh - \sigma(1 - r) > 0,$$

$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z,$$

$$p = \frac{z(1 + xh)E + (1 - r)(1 + xhz)}{2P}, \qquad l = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + ME\}}{2P},$$

$$\kappa = \frac{K(1 + \sigma r)(1 + \sigma z)}{2MP}.$$

The original coefficients are

$$m = \kappa x^2 z H = \kappa x z e, \qquad C = \frac{Kz(1 + \sigma r)E}{2(1 - r)P},$$

$$A = p/r, \quad \alpha = l/(zQ), \quad a = p/b, \quad \ell = l/Q,$$

$$D = \frac{K(1 + \sigma z)(1 + \sigma r) - JP}{2JP} \ge 0, \qquad \beta = \frac{K(1 + \sigma r)(1 + xhz) - \theta P}{2\theta P} \ge 0.$$

Here H has no upper bound. Neither b, Q, θ, $1-r$, nor z is bounded below. The testing rectangle is independently

$$0 \le u, w \le \delta.$$

For signed angles u_s, w_s, set $\alpha_s = \operatorname{sgn}(u_s)$ and $\beta_s = \operatorname{sgn}(w_s)$, choosing the value 1 at zero. The exact testing gauge is

$$(u, w, X, Y, Z, W) = (|u_s|, |w_s|, \beta_s X_s, \alpha_s Y_s, \alpha_s\beta_s Z_s, W_s).$$

It preserves the parent polynomial. Composing every testing function below with this one fixed piecewise semialgebraic map extends the same feature count to $|u_s|, |w_s| \le \delta$; no square root of a negative angle is taken. State coordinates X, Y, Z, W are unrestricted real numbers; no state ratio is bounded.

Define

$$H_0 = 1 + zQ, \quad B_s = P + xz(J - \sigma),$$
$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1 + h + r\theta)] + Ke^2\}.$$

Two exact source identities, obtained by multiplication, are

$$2\theta P\beta = hB_s - \sigma H_0, \tag{1}$$
$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{2}$$

The angular quantities used below are

$$\nu = (2D - 2\beta\theta + h)/\sigma, \quad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z > 0,$$
$$K_A = 4\kappa H - \lambda, \quad d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad S = d_G + \rho,$$
$$L_0 = \lambda\sigma, \quad q_0 = \lambda k, \quad c = \max(0, 1 - K_A), \quad a_0 = k/\sigma.$$

The letter q_0 here is a source quantity, not the constant coefficient of a later scalar channel.

The complete reciprocal parent being certified is

$$\begin{aligned} F_w = {} & 4m(W - wX + uY - uwZ)^2 + 4C(1+w^2)(uW - Y)^2 \\ & + 4A(1+u^2)(1+w^2)Y^2 + (4D+2)(uwW + uX - wY - Z)^2 \\ & + (4\alpha+2)(1+u^2)(Y - wZ)^2 + (4\beta+2)(1+w^2)(X + uZ)^2 \\ & - 2(1+u^2)(1+w^2)Z^2. \end{aligned}$$

All four state coordinates occur; they need not be normalized.

2 Uniform source bounds, without an upper bound on H

All inequalities in this section follow under the larger condition $\varepsilon \le 2^{-22}$, so in particular on the chosen rectangle. Directly from the positive formula for P and the definitions,

$$1 \le P \le 1 + 5\varepsilon, \quad |J-1|, |M-1| \le \varepsilon, \quad 1 \le B_s \le 1 + 7\varepsilon, \quad 1 \le H_0 \le 2.$$

Initially $0 < e < 1$ gives $0 \le \mathcal{H} \le 4$: nonnegativity uses (2). Eliminate β to obtain

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJD/\sigma.$$

Consequently

$$e \le 2x + 4\sigma \le 6\varepsilon, \qquad |\mathcal{H} - H_0| \le 32\varepsilon, \qquad 0 \le D \le 4\sigma^2, \quad 0 \le \beta\theta \le 1. \tag{3}$$

For example, in $\mathcal{H}-H_0$ the terms are bounded by $2\varepsilon+3\varepsilon+3e+2e^2$, which is at most 32ε here. The same positive formulas give

$$|p-H_0/2|\le 16\varepsilon,\quad |l-1/2|\le 8\varepsilon,\quad 1/4\le p,l,\kappa\le 2,\quad \kappa\le 1,\quad m\le 6\varepsilon^2,\quad C(1-r)\le 1. \tag{4}$$

In particular $4a, 4\ell \ge 1$. The unbounded positive coefficients a, ℓ, C will be retained in whole squares and will not be estimated separately.

Exact rearrangement gives

$$\nu = 2D/\sigma + [H_0 - ez(J-\sigma)]/P,$$

$$\lambda = 2H + 2e\theta z(J-\sigma)/P - 2\theta H_0/P + 8Dh/\sigma - 4bp\sigma z.$$

Thus

$$|\nu|\le 3,\qquad |\lambda-2H|\le 5,\qquad |L_0-2h|\le 8\sigma. \tag{5}$$

For completeness, the cancellation needed when H is large is polynomial. Write

$$N_\kappa = K(1+\sigma r)(1+\sigma z) - MP = xF_x + \sigma F_\sigma.$$

Here $F_\sigma = r-1+\sigma rz$, and F_x is the polynomial $(N_\kappa - N_\kappa|_{x=0})/x$. Its coefficient ℓ^1 norm in (x,σ,z,b,h,r) is 37, as can also be checked by expanding the displayed product. Since all six arguments lie in $[0,1]$,

$$H(2\kappa-1) = \frac{eF_x + hF_\sigma}{MP},\qquad |H(2\kappa-1) - h(r-1)|\le 462\varepsilon.$$

The estimate $|MP-1|\le 8\varepsilon$ suffices for the second inequality. Combining this with (5) more precisely yields

$$|K_A - \{2\theta H_0 - 2h(1-r)\}|\le 2^{12}\varepsilon,\qquad |K_A|\le 5,\quad 0\le c\le 6. \tag{6}$$

If $c>0$, then $h\ge 1/8$: for $h<1/8$ the expression in braces is at least $3/2$, whereas the error is less than $1/4$. The earlier elimination identity also gives

$$a_0^2(hB_s - \sigma H_0)\le \mathcal{H}.$$

Therefore, whenever $c>0$,

$$a_0\le 8,\qquad L_0\ge 1/8,\qquad 1/\lambda\le 8\sigma. \tag{7}$$

Every term below containing a_0 or $1/\lambda$ from the angular subtraction is multiplied by c. On $c=0$ that subtraction is identically zero; no uniform estimate on those unused factors is required.

Put

$$R_0 = -4\{2(C+D)(r-1) + 2p + r - 1\}.$$

The positive expression for $C(1-r)$ and the formula for p give $|R_0|\le 256(x+\sigma)\le 512\varepsilon$. Since

$$(x+\sigma)^2 H = xe + 2\sigma e + \sigma h \le 2\varepsilon,$$

(5) further gives

$$|R_0|\sqrt{\lambda}\le 1024\sqrt{\varepsilon}. \tag{8}$$

Finally the exact identity

$$S = \frac{2\mathcal{H}}{P} + d_G(1-J+2hx) + \frac{2eH_0}{P} - \frac{2e^2z(J-\sigma)}{P} - 4bpxz$$

shows

$$|S-2H_0|\le 256\varepsilon,\qquad 1\le S\le 5,\quad 0\le d_G,\rho\le 5,\quad 0<L_0\le 3,\quad 0<q_0\le 4. \tag{9}$$

Here $q_0^2 = L_0\rho$; all bounds use $\lambda>0$ but no lower bound on λ or upper bound on H.

3 A finite formula for the main Gram

The state roots are

$$N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad R = wZ/q - uqW, \quad \Gamma = \sigma Z - uwW,$$
$$A_\theta = q(u^2 - x)W, \quad T_\theta = kq(u - k)W, \quad J_C = N_y + bR, \quad J_\ell = N_y - QR, \quad \Gamma_* = \sqrt{d_G}\Gamma.$$

Let

$$B_1 = \sqrt{\lambda\sigma}\, uZ, \quad B_2 = \sqrt{\rho}\, uwW, \quad B_3 = \sqrt{\lambda}\, xqW, \quad B_4 = \sqrt{\lambda x}\, wZ/q.$$

When $WZ > 0$, put $G_* = \operatorname{sgn}(W)(B_1B_2B_3B_4)^{1/4}$ and $d_i = B_i - G_*$. This is the positive-state convention followed, when necessary, by changing the signs of all roots together. When $WZ \le 0$ put $d_i = B_i$. Equivalently, for every state,

$$G_* = \sqrt{\lambda x}\, \operatorname{sgn}(W)\sqrt{uw\max(WZ, 0)}, \qquad d_i = B_i - G_*.$$

Define

$$U_* = \sqrt{q_0}\sqrt{u}\, q(u - k)W, \qquad K_* = \sqrt{q_0}\sqrt{k}\, q(u - k)W.$$

Use the twelve-root column

$$g = (N_x, J_C, J_\ell, \Gamma_*, A_\theta, T_\theta, d_1, d_2, d_3, d_4, U_*, K_*)^{\mathsf{T}}.$$

The two radial changes are source-only and uniformly bounded:

$$R = J_C - J_\ell, \qquad N_y = QJ_C + bJ_\ell.$$

Set

$$D_* = \operatorname{diag}(2, 1, 1, 1, 1/2, 1/2, 1/2, 1/2, 1/2, 1/2, 1, 1/2).$$

We now define a polynomial quadratic form $\mathcal{E}(g)$, which fully specifies the correction matrix; a cross term is divided by two when placed in its two symmetric entries.

For its coefficients put $t = u^2$, $\xi = w^2$, $g_s = h - Q$ and

$$\begin{aligned}
R_x &= -4\{2Db + (1 + 2m)g_s - 2\beta\theta\}, \quad B_x = 8\kappa e,\\
A_y &= 8(p - l) + 4z\{(1 + 2m)g_s + 2Dh + 2p\sigma\},\\
\chi &= 8(pb + lQ) - 4h + 4eh(2D + 1) - 8Dh\\
&\quad +4z\{4Dbh + 4pb\sigma + (1 + 2m)g_s^2 + 2\beta\theta^2\},\\
k_1 &= 2(1 + 2m)g_s^2 + 8Dhb + 8pb\sigma + 4\beta\theta^2.
\end{aligned}$$

Then

$$\begin{aligned}
E_{yy} &= 2z\{(1 + 2m)u^2 + (1 + 2D)w^2\},\\
E_{RR} &= z\{8bD + 8bp\sigma + 4b\sigma(2C + 2D + 1)(r - 1) + k_1u^2 + 2b^2(2D + 1)w^2\},\\
E_{yR} &= 4z\{u^2[2Dh + 2p\sigma + (1 + 2m)g_s] + b(2D + 1)w^2\\
&\qquad + \sigma(2C + 1)(r - 1) + 2D[1 + \sigma(r - 1)] + 2p\sigma\}.
\end{aligned}$$

The entire first normal is represented by the linear form

$$F_1 = \frac{\sqrt{d_G}}{S}u\Gamma_* + \frac{\sqrt{\rho}}{S}\{kd_1 - ud_2 + (u - k)d_3\} - \frac{\rho}{S}\sqrt{\sigma}T_\theta.$$

The entire second normal is represented by

$$F_2 = \frac{R_0\sqrt{d_G}}{S} qw\Gamma_* + \frac{R_0\rho\sigma z}{S} R - \frac{R_0\sqrt{\rho}}{S}(qw + z\sqrt{\sigma})(d_2 - d_3) + \left\{ \frac{R_0 q_0 z}{S} + \frac{R_0\sqrt{\lambda}\sqrt{\rho}}{S} qw \right\} T_\theta - 8pquwF_1.$$

These are exact identities $F_1 = u\Gamma$ and $F_2 = qw(R_0 - 8pu^2)\Gamma$. In particular they keep both complete state-normal couplings, not only their contact restrictions.

Write

$$\varepsilon_C = 2/\lambda + u + k, \qquad \varepsilon_G = 2a_0 u + (u + k)/L_0.$$

The correction is exactly

$$\begin{aligned}\mathcal{E} = {} & E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR - 4\nu N_xF_1 + J_CF_2 \\ & - 4uqw(2m + 2D + 1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta + \chi uA_\theta R \\ & - \tfrac{1}{2}\rho\sigma R^2 + \sqrt{q_0}\sqrt{u}\,U_*R - \frac{cd_G}{S}\varepsilon_C \sum_{i=1}^{4} d_i^2 - \frac{cd_G}{S}u^2R^2 - \frac{c}{S}\varepsilon_G\Gamma_*^2 \\ & - \frac{c}{S}\{\sqrt{a_0u}\,U_* + \sqrt{a_0k}\,K_*\}^2.\end{aligned}$$

Define G by $g^\mathsf{T}Gg = g^\mathsf{T}D_*g + \mathcal{E}$. This is the actual main Gram after retaining the positive excess of the N_x, J_C, J_ℓ squares, splitting $K_A + c \geq 1$ equally between A_θ, T_θ, and retaining their nonnegative difference. The averaged circuit identity and angular identity used in this step are precisely those of the printed all-H proof at lines 7550–7740. This definition therefore prints every main-Gram entry through one quadratic polynomial instead of twelve rows of expanded rational functions.

4 Rational coefficient bounds and fixed radicals

Put $v = \sqrt{u}$. Every entry of $G - D_*$ is a polynomial in (v, w), with source coefficients, supported on

$$\{1, w, w^2, v, v^2, v^2w, v^4, v^4w\} \subset \{v^\alpha w^\beta : 0 \leq \alpha \leq 4, 0 \leq \beta \leq 2\}.$$

The following elementary bounds provide a reproducible coefficient estimate:

$$|R_x|, |A_y| \leq 32, \quad |\chi| \leq 64, \quad |B_x| \leq 48\varepsilon, \quad |k_1| \leq 16.$$

The coefficients of E_{yy} on (u^2, w^2) are bounded by $(4, 4)$; those of E_{RR} on $(1, u^2, w^2)$ by $(64\sigma, 16, 4)$; those of E_{yR} by $(32\sigma, 12, 8)$. These follow term by term from (3)–(4); in the constant terms use $C(r - 1)$, never C separately.

Here is a single positive-coefficient majorant which verifies all remaining estimates. In it $s = \sqrt{\varepsilon}$, all twelve root letters are formal nonnegative indeterminates, and $n = r' = J_C + J_\ell$. Define

$$f_1 = u\Gamma_* + sd_1 + ud_2 + (u + s)d_3 + sT_\theta,$$

$$\begin{aligned}f_2 = {} & 512s^2w\Gamma_* + 512s^4r' + 512s^2(w + s)(d_2 + d_3) \\ & + (2048s^2 + 1024sw)T_\theta + 16uwf_1.\end{aligned}$$

The absolute coefficient polynomial of $\mathcal{E}$ is bounded by

$$\mathcal{Q} = (4u^2 + 4w^2)n^2 + (64s^2 + 16u^2 + 4w^2)r'^2 + (32s^2 + 12u^2 + 8w^2)nr'$$

$$+ 12N_x f_1 + J_C f_2 + 16uwN_x n + 32uwN_x r' + 48s^2 wN_x A_\theta + 32unA_\theta + 64uA_\theta r'$$
$$+ \tfrac{5}{2}s^2 r'^2 + 2vU_* r' + 6(16s^2 + u + s)\sum_i d_i^2 + 6u^2 r'^2 + (144u + 48s)\Gamma_*^2$$
$$+ 48uU_*^2 + 96vU_* K_* + 48sK_*^2.$$

Indeed (9) bounds $\sqrt{d_G}/S$, $\sqrt{\rho}/S$, d_G/S, ρ/S by one, while (7) gives $\varepsilon_C \le 16\sigma + u + k$ and $\varepsilon_G \le 24u + 8k$ when needed. The two small coefficients in F_2 use $|R_0 q_0| \le 2048\varepsilon$ and (8).

Expansion, or simply evaluation at one of this positive-coefficient polynomial, gives coefficient ℓ^1 norm 8590. Its constant testing part has norm 5030 and every term of that part is divisible by s. Consequently every source coefficient of every nonconstant testing monomial in G has absolute value below 2^{20}, and the source matrix $G(0,0)$ satisfies

$$|G_{ij}(0,0) - (D_*)_{ij}| \le 2^{20}\sqrt{\varepsilon} = 2^{-60}. \tag{10}$$

These estimates hold before any ordering of the tensor product is assumed. They bound pure-source coefficients in their own field.

For $r = 12$ the following exact quadratic-form decomposition uses only fixed directions:

$$G = \sum_i \left(G_{ii} - \frac{11}{48}\right) e_i e_i^{\mathsf{T}} + \sum_{i<j} \frac{1/48 + G_{ij}}{2}(e_i + e_j)(e_i + e_j)^{\mathsf{T}} + \sum_{i<j} \frac{1/48 - G_{ij}}{2}(e_i - e_j)(e_i - e_j)^{\mathsf{T}}.$$

Write each of its 144 scalar channels as

$$q(c,t) = q^{(0)}(c) + \sum_{j=1}^{7} a_j(c) b_j(t),$$

where b_j run through the seven nonconstant monomials in the displayed support. By (10), for every channel

$$q^{(0)} \ge c_0 = 1/256, \qquad |a_j| \le A_* = 2^{30}.$$

In fact the off-diagonal channels have constant term at least $1/96 - 2^{-61}$ and the diagonal ones at least $13/48 - 2^{-60}$. On the independent testing rectangle, $0 \le b_j \le \sqrt{\delta} = 2^{-50} < \gamma$. Hence the exact separated identity

$$q = (q^{(0)} - c_0) + \frac{1}{2}\sum_{j=1}^{7}\big[(A_* + a_j)(\gamma + b_j) + (A_* - a_j)(\gamma - b_j)\big]$$

has nonnegative factors in their own labelled fields. It proves the required tensor SOS without an appeal to entrywise positivity in a tensor ring or to a point-dependent real congruence.

All twelve roots are source-linear combinations of the fixed testing functions

$$\mathcal{V} = \{X, Y, Z, W, uZ, wZ, uW, uwW, u^2 W, u^{3/2} W, u^{1/2} W, \operatorname{sgn}(W)\sqrt{uw\max(WZ, 0)}\}.$$

The last function includes both state-sign cases and is zero when $WZ \le 0$. Changing all state signs changes all twelve roots' signs and preserves the quadratic form. Therefore a fixed main-Gram dictionary is

$$\mathcal{V} \cup \{f\sqrt{\gamma + b_j}, f\sqrt{\gamma - b_j} : f \in \mathcal{V},\ 1 \le j \le 7\}.$$

It has 180 displayed functions. All radicals are testing-only; the remaining coefficients are source-only square roots. The retained positive squares are included explicitly below.

5 The two zero faces are present, and H is unrestricted

No inverse of D, β, b, Q or a state coordinate was used in the coefficient estimates or Gram decomposition. At $D = 0$ the root Γ_* is zero and all formulas remain literal. At $\beta = 0$ the same formulas apply. Testing axes are included. The denominators S, M, P, J, σ, k used above are nonzero on the stated source domain; their potentially dangerous combinations were bounded explicitly rather than omitted on a boundary.

For explicit nonemptiness set $x = \sigma = t$, $z = 1$, $b = 1/2$, $h = Ht$, with $0 < t \le \varepsilon$. The numerators $D_N = 2JPD$ and $\beta_N = 2\theta P\beta$ are exactly

$$\begin{aligned} D_N &= (3 - H)t^2 + (3/4 - 6H)t^3 + (H^2 + H/2)t^4 \\ &\quad + (-7H^2/2 + H/4)t^5 + (H^3 - H^2/2)t^6, \\ \beta_N &= (H - 3/2)t + 4Ht^2 - Ht^3 + 4H^2t^4 + (-H^3 + H^2/2)t^5. \end{aligned}$$

Uniformly for $1 \le H \le 4$, the remainders beyond the first terms have absolute values at most $(687/4)t^3$ and $156t^2$. Thus continuity gives a $\beta = 0$ source with $H \in [1, 2]$ and $D > 0$, and a $D = 0$ source with $H \in [2, 4]$ and $\beta > 0$. On the first face $p \le 7/8$ and $\lambda = 8DH + 2(H^2 - p)t \ge t/4$. On the second, $\beta\theta \ge t/16$, so $\lambda \ge 1/8 - 4t > 0$. All strict native gaps hold.

Finally take $\sigma = t$, $x = t^2/16$, $z = 1$, $b = 1/2$, $h = 3/4$. Then $H = 3/(4t)$ is unbounded along $t \downarrow 0$ and

$$D_N = \frac{-3t^2}{262144}(6t^5 - 61t^4 + 608t^3 - 7872t^2 + 8192t - 126976) > 0,$$

$$\beta_N = \frac{3(3t^2 + 64)(2t^3 + 5t^2 + 64)}{16384} > 0.$$

Also $K_A = O(\varepsilon)$ on this path by (6), so $c > 0$ is represented. These are singular limits inside the evaluated rectangle, not merely a nonsingular compact slice.

6 Fixed testing functions for the complete positive-angular parent

Retain the preceding source/testing rectangle and $\lambda > 0$. Compose the testing functions with the angle gauge already defined, then multiply all four state coordinates by $\mathrm{sgn}_+(W)$, where $\mathrm{sgn}_+(0) = 1$. Thus $W \ge 0$. Put $Z_+ = \max(Z, 0)$ and $Z_- = \max(-Z, 0)$. The common state sign preserves the quadratic parent. Every testing function below is pulled back through this one fixed map.

6.1 The exact whole-parent decomposition

All symbols $g, G, N_x, N_y, R, \Gamma, A_\theta, T_\theta, J_C, J_\ell$ retain their preceding definitions. Set $F_A = u(Z - a_0 wW)$, $X_e = X + ewW$, and

$$a_C = cd_G/S, \quad q_C^{(0)} = \tfrac{1}{2} - a_C(2/\lambda + k), \quad p_C = q_C^{(0)} - a_C\delta.$$

The proved bounds give $0 \le a_C \le 6$ and $p_C > 1/4$. The circuit remainder satisfies

$$\mathscr{C} = \sum_{i=1}^{4} d_i^2 + \mathscr{C}_+,$$

and the angular difference is

$$\Delta_A = (2/\lambda + u + k)\mathscr{C} + u^2R^2 + \{2a_0u + (u + k)/L_0\}\Gamma^2 - A_\theta^2.$$

The complete parent is exactly

$$\begin{aligned}F_w &= g^{\mathsf{T}}Gg + \mathcal{E}_+ + (2+4m)w^2N_x^2 + \{4a + 2z(2C+1) - 1\}J_C^2 + (4\ell - 1)J_\ell^2 \\ &\quad + \tfrac{1}{2}L_0F_A^2 + \frac{K_A + c - 1}{2}(A_\theta^2 + T_\theta^2) + \frac{K_A + c}{2}(A_\theta^2 - T_\theta^2) \\ &\quad + \frac{cd_G}{S}\Delta_A + \{p_C + a_C(\delta - u)\}\mathscr{C}_+,\end{aligned}$$

where

$$\begin{aligned}\mathcal{E}_+ &= 4\beta(1+w^2)N_x^2 + 2(2D+1)(uX_e)^2 + 4bpzu^2\Gamma^2 \\ &\quad + 4a(u^2 + w^2 + u^2w^2)J_C^2 + 4\ell u^2J_\ell^2 + 4Czw^2J_C^2.\end{aligned}$$

After the expansions below, every square coefficient outside G is nonnegative and source-only, except the explicitly separated testing factor $\delta - u$, which is nonnegative on the fixed testing rectangle. This equality is the full-state identity, not its restriction to a stationary state or a determinant. No inverse of D or β occurs.

6.2 Ordinary retained squares: seven additional functions

Let $\mathcal{V}$ be the twelve pre-Gram functions

$$\{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^{3/2}W, u^{1/2}W, \sqrt{uwWZ_+}\}.$$

All ordinary retained roots are source-linear combinations of $\mathcal{V}$ and the seven functions

$$wX,\ uwZ,\ uX,\ u^2wW,\ uY,\ wY,\ uwY.$$

Here are all seventeen roots; every displayed square root of a coefficient is source-only:

family	square roots
$\mathcal{E}_+$	$2\sqrt{\beta}N_x,\quad 2\sqrt{\beta}\,wN_x,\quad \sqrt{2(2D+1)}\,uX_e,\quad 2\sqrt{bpz}\,u\Gamma$
	$2\sqrt{a}\,uJ_C,\quad 2\sqrt{a}\,wJ_C,\quad 2\sqrt{a}\,uwJ_C,\quad 2\sqrt{\ell}\,uJ_\ell,\quad 2\sqrt{Cz}\,wJ_C$
N_x excess	$\sqrt{2+4m}\,wN_x$
radial excess	$\sqrt{4a + 2z(2C+1) - 1}\,J_C,\quad \sqrt{4\ell - 1}\,J_\ell$
averaged square	$\sqrt{L_0/2}\,F_A$
angular excess	$\sqrt{(K_A + c - 1)/2}\,A_\theta,\quad \sqrt{(K_A + c - 1)/2}\,T_\theta$
angular difference	$\sqrt{(K_A + c)/2}\,qu(u-k)W,$
	$\sqrt{(K_A + c)/2}\,q\sqrt{2k}\sqrt{u}(u-k)W.$

For example $J_C = Y/q - bquW$ and $J_\ell = Y/q - wZ/q + QquW$, so multiplying by u, w, uw adds exactly the stated functions. All other rows follow by substitution. The last two squares use $A_\theta^2 - T_\theta^2 = z(u-k)^2W^2(u^2 + 2uk)$.

6.3 The angular difference: nineteen functions

The following calculation clears the intermediate state ratio completely. Define positive source constants

$$A_u = 1/k,\quad A_v = 1/(\sqrt{\sigma}q),\quad A_z = \sqrt{\sigma}/(kq),\quad M_0 = k^2q,\quad a_* = \lambda k,$$

and write $r_* = \sqrt{wW|Z|}$. Eight cleared roots are

$$\begin{aligned}
f_1 &= M_0(A_uA_vA_zuw|Z| - W),\\
f_2 &= \sqrt{2}M_0(A_vA_zw|Z| - A_uuW),\\
f_3 &= M_0\{(\sqrt{a_*} + 1/\sqrt{a_*})A_uA_vuwW - \sqrt{a_*}A_uA_zu|Z| - A_z|Z|/\sqrt{a_*}\},\\
f_4 &= \sqrt{A_u}\sqrt{u}\, f_3,\\
f_5 &= \sqrt{a_*}M_0(W - A_vA_zw|Z|), \qquad f_6 = \sqrt{A_u}\sqrt{u}\, f_5,\\
f_7 &= \sqrt{2a_*A_vA_z}M_0(A_uu - 1)r_*, \qquad f_8 = \sqrt{A_u}\sqrt{u}\, f_7.
\end{aligned}$$

For $Z \geq 0$, $\Delta_A = \sum_{i=1}^{8} f_i^2$. To verify this, put $U = A_uu$, $V = A_vw$, $Z_* = A_z|Z|/W$, $M = M_0W$; the roots are M times the eight roots obtained by expanding every $U + 1$ factor in the identity

$$\begin{aligned}
&[2 + a_*(U + 1)]\Psi + U^2Q_*^2 + [2U + (U + 1)/a_*]R_*^2 - T_*^2\\
&= (T_* + UQ_*)^2 + 2Q_*^2 + (U + 1)(\sqrt{a_*}U(V - Z_*) - R_*/\sqrt{a_*})^2\\
&\quad + a_*(U + 1)(1 - VZ_*)^2 + 2a_*(U + 1)VZ_*(U - 1)^2,
\end{aligned}$$

where $\Psi = 1 + U^2V^2 + U^2Z_*^2 + V^2Z_*^2 - 4UVZ_*$, $Q_* = VZ_* - U$, $R_* = Z_* - UV$, and $T_* = U^2 - 1$. It is a polynomial identity by multiplication.

For $Z < 0$, add three squares. Put

$$C_2 = 4, \quad C_1 = 8a_* + 8 + 4/a_*, \quad C_0 = 16 + 8a_* + 4/a_*.$$

Their roots are

$$M_0\sqrt{A_uA_vA_z}\,\mathbf{1}_{Z<0}\big(\sqrt{C_2}A_uu^{3/2}r_*, \quad \sqrt{C_1A_u}\,ur_*, \quad \sqrt{C_0}\sqrt{u}\,r_*\big).$$

Indeed changing Z_* to $-Z_*$ increases the left side by $M^2UV|Z_*|(C_2U^2 + C_1U + C_0)$. This also proves the exact negative-state-sector coefficients. Every displayed source factor is a real closed field square root in the source factor.

The resulting nineteen testing functions, including overlaps with $\mathcal{V}$, are

$$\begin{aligned}
\mathcal{A} = \{&W, uw|Z|, w|Z|, uW, uwW, u|Z|, |Z|, u^{3/2}wW, u^{3/2}|Z|, \sqrt{u}|Z|, \sqrt{u}W, \sqrt{u}w|Z|,\\
&ur_*, r_*, u^{3/2}r_*, \sqrt{u}r_*, \mathbf{1}_{Z<0}u^{3/2}r_*, \mathbf{1}_{Z<0}ur_*, \mathbf{1}_{Z<0}\sqrt{u}r_*\}.
\end{aligned}$$

Multiply the eleven root formulas above by the pure-source factor $\sqrt{cd_G/S}$ for their contribution to F_w. Every cleared formula is defined at $W = 0$ and $Z = 0$. After substituting squares for $u, w, W, |Z|$, the identities are polynomial; therefore they include those literal boundaries, without a limit or a state-coordinate inverse in the final formulas.

6.4 The circuit remainder: fourteen functions

Set the positive source constants

$$s_1 = \sqrt{\lambda\sigma}, \quad s_2 = \sqrt{\rho}, \quad s_3 = \sqrt{\lambda}\,xq, \quad s_4 = \sqrt{\lambda x}/q, \qquad g_0 = \sqrt{\lambda x}.$$

Define seven testing functions

$$\begin{aligned}
t_1 &= u^{3/4}w^{1/4}W^{1/4}Z_+^{3/4}, \quad t_2 = u^{3/4}w^{3/4}W^{3/4}Z_+^{1/4},\\
t_3 &= u^{1/4}w^{1/4}W^{3/4}Z_+^{1/4}, \quad t_4 = u^{1/4}w^{3/4}W^{1/4}Z_+^{3/4},\\
t_5 &= u^{3/4}w^{1/2}W^{1/2}Z_+^{1/2}, \quad t_6 = u^{1/4}w^{1/2}W^{1/2}Z_+^{1/2},\\
t_7 &= \sqrt{uwWZ_-}.
\end{aligned}$$

An explicit full-sector SOS is

$$\mathscr{C}_+ = r_1^2 + r_2^2 + r_3^2 + r_4^2,$$
$$r_1 = \sqrt{2g_0}(\sqrt{s_1}t_1 - \sqrt{s_2}t_2), \quad r_2 = \sqrt{2g_0}(\sqrt{s_3}t_3 - \sqrt{s_4}t_4),$$
$$r_3 = 2\sqrt{g_0}\{(s_1s_2)^{1/4}t_5 - (s_3s_4)^{1/4}t_6\}, \qquad r_4 = 2\sqrt{\lambda x}\, t_7.$$

The first three roots vanish for $Z \le 0$ and the last vanishes for $Z \ge 0$. They also vanish wherever $WZ = 0$. On $Z > 0$ they are precisely the separated two-pairing four-monomial identity; on $Z < 0$ the remainder is $4\lambda xuwWZ_-$.

There is no point-dependent scalar reserve at this step:

$$(p_C + a_C(\delta - u))\mathscr{C}_+ = \sum_{j=1}^{4}(\sqrt{p_C}r_j)^2 + \sum_{j=1}^{4}(\sqrt{a_C}\sqrt{\delta - u}\, r_j)^2.$$

Thus the fixed circuit feature list is exactly the seven t_i and their seven products with $\sqrt{\delta - u}$.

6.5 The complete testing space

For the main Gram take the seven nonconstant monomials

$$\mathcal{B} = \{w, w^2, \sqrt{u}, u, uw, u^2, u^2w\}, \qquad \gamma_7 = \frac{1}{7\,2^{38}}.$$

The main list is $\mathcal{V}$ times the fifteen fixed factors $1, \sqrt{\gamma_7 + b}, \sqrt{\gamma_7 - b}$ for $b \in \mathcal{B}$, giving 180 functions. Its 144 fixed directions are e_i, $e_i + e_j$, and $e_i - e_j$, with the exact scalar-channel formulas derived above, using $7A_*\gamma_7 = c_0$.

Theorem 1. *On the stated independent rectangle, the full real-state parent F_w is a sum of squares of source-linear combinations of one fixed list of* 220 *labelled testing functions: the* 180 *main functions, the seven ordinary functions, the nineteen angular functions, and the fourteen circuit functions. Repeated functions are harmless; the real dimension of their span is at most* 220*. There is an explicit construction using at most* 2196 *square roots:* $144 \cdot 15$ *main roots, seventeen ordinary roots, eleven angular roots, and eight circuit roots. The same list, composed with the fixed gauges above, includes all real state-sign sectors, both testing axes, and the source faces $D = 0$ and $\beta = 0$.*

This theorem follows from the displayed exact decomposition and the separated coefficient formulas. It needs no Gram factorization with mixed radicals. Without using the physical reality reduction, apply the theorem separately to the real and imaginary state amplitudes; their union has at most 440 functions and 4392 roots. The parent is real quadratic in the state, so those two contributions sum to its complex-state value.

7 The complete nonpositive-angular branch

The same source identities also control $\lambda \le 0$. In this case, a source-only shear replaces the angular circuit and gives a fixed positive diagonal Gram matrix.

7.1 Domain and source bounds

Keep the same rational source formulas and strict gaps, and now assume $\lambda \le 0$. Thus

$$0 < x, \sigma \le \varepsilon = 2^{-160}, \quad 0 < z \le 1, \quad 0 < b < 1, \quad r = bz < 1, \quad 0 < h < 1, \quad D, \beta \ge 0,$$
$$0 \le u, w \le \delta = 2^{-100}.$$

State coordinates W, X, Y, Z are unrestricted. Set $k = \sqrt{x}$, $q = \sqrt{z}$, $Q = 1-b$, $H = h/\sigma$, $H_0 = 1+zQ$, $d_G = 4D/\sigma^2$, $L = \lambda/\sigma$, and $K_c = 1 + L^2$. The source identities BE3–BE4 give the following finite bounds:

$$H \le 2, \quad h \le 2\sigma, \quad -8 \le L \le 0, \quad |d_G - 2H_0| \le 200\varepsilon, \quad |4\kappa H - 2H_0| \le 400\varepsilon. \tag{N1}$$

The already established common bounds used below are

$$1 \le H_0 \le 2, \quad |\nu| \le 3, \quad p \le 2, \quad m \le 6\varepsilon^2, \quad D \le 4\sigma^2, \quad |R_0| \le 512\varepsilon, \quad |B_x| \le 48\varepsilon,$$

$$|R_x|, |A_y| \le 32, \quad |\chi| \le 64, \quad |k_1| \le 16, \qquad 4a, 4\ell \ge 1. \tag{N2}$$

Here all coefficient symbols have their displayed definitions in the preceding main-Gram construction, and $R_\gamma = R_0 - 8pu^2$. The coefficient bounds for E_{yy}, E_{RR}, E_{yR} on their respective testing supports (u^2, w^2) and $(1, u^2, w^2)$ are

$$(4, 4), \qquad (64\sigma, 16, 4), \qquad (32\sigma, 12, 8). \tag{N3}$$

These bounds retain $C(1 - r)$ rather than estimating the possibly unbounded positive coefficient C. Their source-formula proofs in the shared calculation do not use $\lambda > 0$; only the subsequent positive-angular construction there uses that sign.

Here is a short derivation of (N1), using only the shared source formulas and common bounds. Multiplying $\lambda \le 0$ by σ gives

$$8Dh + 4\beta\theta^2 + 2h^2 \le 4bp\sigma^2 z.$$

Thus $H^2 \le 2bpz \le 4$, $h \le 2\sigma$, and $\beta\theta^2/\sigma^2 \le bpz \le 2$. In particular $\theta \ge 1 - 2\varepsilon$ and $\beta\theta/\sigma^2 \le 3$. Also every positive term in L can be discarded to give $L \ge -4bpz \ge -8$. For the remaining estimates it is enough to use $\varepsilon \le 1/2000$. The positive formula for P, with $h \le 2\sigma$, yields $1 \le P \le 1 + 4\varepsilon$, and $1 \le B_s \le 1 + 6\varepsilon$. BE3 is exactly

$$HB_s - H_0 = 2P\beta\theta/\sigma.$$

Consequently

$$|H - H_0| \le \left\{ \frac{4(1 + 4\varepsilon)}{1 - 2\varepsilon} + 12 \right\} \varepsilon < 18\varepsilon.$$

Use the common bound $|\mathcal{H} - H_0| \le 32\varepsilon$ and $|PJ - 1| \le 6\varepsilon$, $PJ \ge 1 - \varepsilon$. BE4 gives

$$d_G = \frac{2\mathcal{H}}{PJ} - \frac{4x\beta\theta}{J\sigma^2}.$$

The first term differs from $2H_0$ by at most $88\varepsilon/(1 - \varepsilon)$ and the second has absolute value at most $12\varepsilon/(1 - \varepsilon)$, so their sum is less than 200ε. Finally $|M - 1| \le \varepsilon$ implies $|MP - 1| \le 6\varepsilon$, while $1 \le K(1 + \sigma r)(1 + \sigma z) \le 1 + 4\varepsilon$. The formula for κ therefore gives $|\kappa - 1/2| \le 5\varepsilon/(1 - \varepsilon) < 39\varepsilon$. Together with $H \le 2$, this proves

$$|4\kappa H - 2H_0| \le 8(39\varepsilon) + 2(18\varepsilon) < 400\varepsilon.$$

7.2 Exact complete quadratic correction

Retain the uniform proof's roots

$$N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad R = wZ/q - uqW, \quad \Gamma = \sigma Z - uwW,$$
$$A_\theta = q(u^2 - x)W, \quad V = uw(u^2 - x)W, \quad J_C = N_y + bR, \quad J_\ell = N_y - QR, \quad X_e = X + ewW.$$

In particular $R = J_C - J_\ell$ and $N_y = QJ_C + bJ_\ell$. Introduce the source-only shear

$$T = \Gamma + (L/2)V, \qquad g = (N_x, J_C, J_\ell, T, V, A_\theta)^{\mathsf{T}}, \qquad D_* = \mathrm{diag}(2,1,1,1,1,2).$$

This shear is invertible with inverse $\Gamma = T - (L/2)V$; both directions have coefficients bounded by four. The source baseline has the exact decomposition

$$\begin{aligned} 2H_0\Gamma^2 + L\Gamma V + (1+L^2)V^2 &= T^2 + V^2 + (2H_0-1)\Gamma^2 + \tfrac{3}{4}L^2V^2, \\ 2H_0A_\theta^2 &= 2A_\theta^2 + 2(H_0-1)A_\theta^2. \end{aligned} \tag{N4}$$

Thus the principal baseline is the fixed diagonal D_*, with the last terms in (N4) retained as nonnegative source squares. This is a source-only change of roots, not a congruence depending on testing parameters. Below all final directions are fixed in these six coordinates.

Define $\mathcal{E}$ by the following quadratic polynomial, substituting $\Gamma = T - (L/2)V$, $R = J_C - J_\ell$, and $N_y = QJ_C + bJ_\ell$:

$$\begin{aligned} \mathcal{E} = {} & E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR + (d_G - 2H_0)\Gamma^2 - 4\nu uN_x\Gamma + qw(R_0 - 8pu^2)J_C\Gamma \\ & - 4uqw(2m+2D+1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta \\ & + (\chi+\lambda)uA_\theta R + (4\kappa H - 2H_0)A_\theta^2 + Lu^2\Gamma^2 \\ & - K_c\sigma u^2A_\theta^2 - K_c\sigma u(u^2-x)A_\theta R + K_c(u^2-x)V\Gamma. \end{aligned} \tag{N5}$$

This specifies every entry of G by $g^{\mathsf{T}}Gg = g^{\mathsf{T}}D_*g + \mathcal{E}$; cross coefficients are divided by two between symmetric entries.

For completeness the entire retained part is

$$\begin{aligned} \Pi = {} & 4\beta N_x^2 + (4\beta + 2 + 4m)(wN_x)^2 + 2(2D+1)(uX_e)^2 + 4bpz(u\Gamma)^2 \\ & + 4a\{(uJ_C)^2 + (wJ_C)^2 + (uwJ_C)^2\} + 4\ell(uJ_\ell)^2 + 4Cz(wJ_C)^2 \\ & + [4a + 2z(2C+1) - 1]J_C^2 + (4\ell - 1)J_\ell^2 \\ & + (2H_0-1)\Gamma^2 + \tfrac{3}{4}L^2V^2 + 2(H_0-1)A_\theta^2. \end{aligned} \tag{N6}$$

Every coefficient in (N6) is nonnegative in the source field. No mixed radical occurs.

The exact parent identity is

$$F_w = \Pi + g^{\mathsf{T}}Gg. \tag{N7}$$

Indeed the uniform proof gives

$$K_AA_\theta^2 + \mathscr{S} = 4\kappa HA_\theta^2 + Lu^2\Gamma^2 + L\Gamma V + \lambda uA_\theta R.$$

Add K_c times its exact zero identity

$$V^2 - \sigma u^2A_\theta^2 - u\sigma(u^2-x)A_\theta R + (u^2-x)V\Gamma = 0.$$

Splitting the radial and N_x pivots and then using (N4) gives precisely (N5)–(N7), including both complete Gamma normal terms.

7.3 Evaluated coefficient bound and fixed directions

All entries of $G - D_*$ are polynomials on the fixed testing support

$$\{1, w, w^2, u, uw, u^2, u^2w, u^3\}. \tag{N8}$$

Here is a positive-coefficient majorant for the absolute coefficient polynomial of (N5). Treat its root letters as formal nonnegative variables, put $n = J_C + J_\ell$, and $C_* = T + 4V$. Then

$$\begin{aligned}\mathcal{Q} = {} & (4u^2+4w^2)n^2 + (64\varepsilon + 16u^2 + 4w^2)n^2 + (32\varepsilon + 12u^2 + 8w^2)n^2 \\ & + 200\varepsilon C_*^2 + 12uN_xC_* + (512\varepsilon w + 16u^2 w)J_C C_* \\ & + 16uwN_x n + 32uwN_x n + 48\varepsilon wN_x A_\theta + 32unA_\theta + 65uA_\theta n \\ & + 400\varepsilon A_\theta^2 + 8u^2C_*^2 + 65\varepsilon u^2 A_\theta^2 + 65\varepsilon(u^3+u)A_\theta n + 65(u^2+\varepsilon)VC_*.\end{aligned} \tag{N9}$$

For the two replacement terms use $|\lambda| \le 8\varepsilon$, $|\chi + \lambda| \le 65$, $K_c \le 65$, and $x, \sigma \le \varepsilon$. All other terms use (N1)–(N3). In particular $|2m + 2D + 1| \le 4$, a deliberately loose bound.

Evaluating this positive polynomial at one gives total coefficient norm $10189 < 2^{14}$. Its constant-testing part has norm 6109ε. The successive summands contribute

$$32, 336, 208, 5000, 60, 2640, 32, 64, 48, 64, 130, 400, 200, 65, 260, 650$$

to the first total. The constant part consists of $(256 + 128 + 5000 + 400 + 325)\varepsilon$. Consequently the deliberately larger rational bounds

$$|G_{ij}(0,0) - (D_*)_{ij}| \le 2^{20}\varepsilon, \quad \text{every nonconstant coefficient has magnitude } \le 2^{20} \tag{N10}$$

hold in the source field.

For six coordinates use the exact fixed-direction decomposition

$$\begin{aligned}G = {} & \sum_i (G_{ii} - 5/16)e_i e_i^{\mathsf{T}} \\ & + \sum_{i<j} \frac{1/16 + G_{ij}}{2}(e_i + e_j)(e_i + e_j)^{\mathsf{T}} + \sum_{i<j} \frac{1/16 - G_{ij}}{2}(e_i - e_j)(e_i - e_j)^{\mathsf{T}}.\end{aligned} \tag{N11}$$

There are 36 scalar channels. Let $b_1, \dots, b_7$ be the nonconstant monomials of (N8), fix

$$c_0 = 2^{-8}, \qquad A_* = 2^{30}, \qquad \gamma = \frac{1}{7\,2^{38}},$$

and write a channel as $q_0(c) + \sum_j a_j(c)b_j(t)$. The diagonal constant channels are at least $11/16 - 2^{20}\varepsilon$; the off-diagonal ones are at least $1/32 - 2^{19}\varepsilon$. Thus $q_0 \ge c_0$ and $|a_j| \le A_*$. Also $0 \le b_j \le \delta < \gamma$ throughout the independent testing box. The exact separated identity

$$q_0 + \sum_j a_j b_j = (q_0 - c_0) + \frac{1}{2}\sum_{j=1}^{7}\{(A_* + a_j)(\gamma + b_j) + (A_* - a_j)(\gamma - b_j)\} \tag{N12}$$

uses $7A_*\gamma = c_0$. Every factor on its right is nonnegative in its own labelled field. Multiplying these squares by the fixed linear directions in (N11) proves (N7) by an evaluated finite tensor sum of squares, without an infinitesimal-residue argument on a finite rectangle.

7.4 A fixed complete testing space

Every coordinate of g is source-linearly spanned by

$$\mathcal{V}_0 = \{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^3wW\}. \tag{N13}$$

For example $J_C = Y/q - buqW$, $J_\ell = Y/q - wZ/q + QuqW$, and $T = \sigma Z - uwW + (L/2)(u^3wW - xuwW)$. Thus the main Gram uses the 150 displayed functions

$$\mathcal{V}_0 \ \cup \ \{f\sqrt{\gamma + b_j},\ f\sqrt{\gamma - b_j} : f \in \mathcal{V}_0,\ 1 \le j \le 7\}. \tag{N14}$$

All retained roots in (N6) lie in the source-linear span of $\mathcal{V}_0$ together with the seven additional functions

$$\{wX, uwZ, uX, u^2wW, uY, wY, uwY\}. \tag{N15}$$

For instance $uX_e = uX + euwW$ and $u\Gamma = \sigma uZ - u^2wW$. This proves a complete-parent dictionary of 157 displayed testing functions. Each source-only linear combination obtained from (N11)–(N12) and (N6) has a nonnegative source coefficient, absorbed by its own source square root. The formulas give the finite coefficient prescription; no successful global pencil or global feature bound follows.

For signed testing angles compose the entire dictionary with the source-independent angle gauge defined above; this does not change the count. The polynomial identities specialize literally at $u = 0$, $w = 0$, and arbitrary zero state coordinates. Applying the same real certificate to the real and imaginary parts treats complex states (with the corresponding doubled real dictionary).

No inverse of D or β appears. The $\beta = 0$ face is allowed literally. In this branch (N1) actually implies $d_G \ge 2 - 200\varepsilon > 0$, so the $D = 0$ face has empty intersection with the stated box and $\lambda \le 0$. Actual $D = 0$ sources are therefore supplied by the positive-λ branch, not by this note. This distinction is a consequence of the explicit bounds, not an additional exclusion.

Nonempty sources, including the zero-beta face

For a completely explicit source path set

$$x = \sigma = t, \qquad z = b = 1 - t^2, \qquad h = t - 4t^2 + Bt^3, \qquad 16 \le B \le 18, \quad 0 < t \le \varepsilon.$$

All strict native gaps hold, since $0 < h \le 2t$, $E = 1 - h > 0$, and $J, M \ge 1 - t > 0$. Let $p_N = 2Pp$, $D_N = 2JPD$, $\beta_N = 2\theta P\beta$, and

$$N_L = 4D_N h + 2\beta_N \theta J + 2h^2 JP - 2bp_N\sigma^2 zJ = \sigma^2 JPL.$$

Direct polynomial multiplication, grouping by powers of t, gives

$$\begin{aligned} |D_N - t^2| &\le 27640t^3, \\ |\beta_N - (B-17)t^3| &\le 24459t^4, \\ |N_L - (2B-38)t^3| &\le 248856t^4. \end{aligned}$$

For these bounds take absolute coefficient sums in (t, B) and replace every B^j by 18^j in the remainders. Thus $D > 0$ and $L < 0$ throughout this path. At $B = 18$, also $\beta > 0$, giving an entirely rational strict source for each rational t. The signs of β_N at $B = 16, 18$ are opposite. Polynomial continuity therefore gives $B \in (16, 18)$ with $\beta = 0$, while $D > 0$ and $L < 0$ remain strict. This verifies actual zero-beta sources in the branch; it is not only a formal boundary allowance.